

Jingtao Qin

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EDUCATION

University of California, Riverside

Ph.D. of Electrical and Computer Engineering, GPA:3.97/4

Riverside, CA

January 2021 – July 2025

Shandong University

M.S. of Electrical Engineering

Jinan, China

September 2018 – June 2020

B.E. of Electrical Engineering, GPA: 4.51/5, Rank: 12/277

September 2014 – June 2018

EXPERIENCE

Power System Research Scientist

Hitachi America, Ltd.

July 2025 – Present

Santa Clara, CA

- Conduct research in mathematical optimization for power system analysis, focusing on both distribution and transmission.
- Collaborate with internal and external cross-functional teams, including utility partners and academic institutions
- Publish research findings in peer-reviewed paper/journal.

Research Assistant

Smart City Innovation Laboratory

January 2021 – July 2025

UC Riverside

- Developed an optimization method-assisted ensemble deep reinforcement learning algorithm to accelerate Unit Commitment (UC) solutions in the day-ahead electricity market.
- Developed a hierarchical graph convolution network (GCN) model including spatio-temporal graph convolution network (STGCN) and edge-conditioned graph network (ECGCN) to accelerate Unit Commitment (UC) solutions in the day-ahead electricity market.
- Co-developed a power system voltage health index prediction model using the graph convolutional network (GCN).

DERMS Software Intern

Mitsubishi Electric Research Laboratories, Inc.

June 2024 – September 2024

Cambridge, MA

- Developed a stochastic dynamic hosting capacity assessment module for Distributed Energy Resource Management System(DERMS) using temporal Surrogate Lagrangian Decomposition.

Data Analyst Intern

National Renewable Energy Laboratory

June 2023 – September 2023

Denver, CO

- Developed a physics-informed spatio-temporal graph convolution network for dynamic reconfiguration problems to minimize the curtailment of solar generations in unbalanced distribution systems.

COURSES (SELECTED)

Pattern Recognition|Introduction to Deep Learning|Optimization for Machine Learning|Introduction to Power Distribution Systems|Power System Steady State and Market Analysis|Mathematical Methods for Electrical Engineers|Convex Optimization

TECHNICAL SKILLS

Python, SQL, OpenDSS|PSS/E, py-dss-interface, Pytorch, Scikit-learn, Pandas, PyTorch Geometric, Pyomo|Gurobi|Cplex|SCIP

PUBLICATIONS (SELECTED)

- **Jingtao Qin**, Anbang Liu, Mikhail Bargin, Nanpeng Yu. "AccSPS Learning Rate: Accelerated Convergence through Decision-Adjusted Levels for Stochastic Polyak Stepsize." *IEEE Transactions on Automation Science and Engineering*, Early Access, 2025.
- **Jingtao Qin**, Nanpeng Yu, "Solve Large-scale Unit Commitment Problems by Physics-informed Graph Learning," *IEEE Transactions on Power Systems*, 2025.
- **Jingtao Qin**, Rui Yang, Nanpeng Yu. "Physics-informed Graph Neural Networks for Collaborative Dynamic Reconfiguration and Voltage Regulation in Unbalanced Distribution Systems," *IEEE Transactions on Industry Applications*, 2025.

- **Jingtao Qin**, Yuanqi Gao, Mikhail Bragin, and Nanpeng Yu, "An Optimization Method-Assisted Ensemble Deep Reinforcement Learning Algorithm to Solve Unit Commitment Problems," *IEEE Access*, 2023.
- **Jingtao Qin**, Hongbo Sun, Jianlin Guo, Ye Wang, Arvind Raghunathan, Nanpeng Yu. "Temporal Surrogate Lagrangian Decomposition for Stochastic Dynamic Hosting Capacity Assessment In Unbalanced Distribution Systems." *IEEE Transactions on Industrial Informatics*, 2024, under the second round of review.
- **Jingtao, Qin**, Nanpeng Yu, and Yuanqi Gao. "Solving unit commitment problems with multi-step deep reinforcement learning." *IEEE SmartGridComm*, 2021 .
- **Jingtao Qin** and Nanpeng Yu, "Reconfigure Distribution Network with Physics-informed Graph Neural Network," *IEEE ISGT Europe*, 2023.
- **Jingtao Qin**, Xueshan Han, Guojing Liu, Shang Wang, Wenbo Li, Zhe Jiang, "Wind and storage cooperative scheduling strategy based on deep reinforcement learning algorithm," *Journal of Physics: Conference Series*, 2019.

CERTIFICATE

- Optimization 201/202 for Data Scientists (@Gurobi)

HONORS AND SCHOLARSHIPS (SELECTED)

- **Dissertation Completion Fellowship Award (DCFA)**. Awarded by UC Riverside, 2024.
- **Dean's Distinguished Fellowship**. Awarded by UC Riverside, 2021.